Turtle Games

Predicting Future Outcomes

Predicting Future Outcomes

Background

Turtle Games is a game manufactureer and retailer with a global customer base. The company manufactures and sells its own products, along with sourcing and selling products manufactured by other companies. Its product range includes books, board games, video games, and toys.

Turtle Games provided sales and customer review data to be analysed, with the primary business objective of improving overall sales performance. Turtle Games would like to understand how customers accumulate 'loyalty points', how groups within the customer base can be used to target specific market segments and how customer reviews can be used to inform marketing campaigns.

Exploratory Data Analysis

Turtle Games provided one data file which included both qualitative and quantitative data for 2000 customers split over 9 columns. Comprehensive exploratory analysis was conducted using the provided dataset in Python and R, focusing on customer segmentation, loyalty points and customer reviews. The initial review highlighted two redundant columns: language and platform, which were removed.

Of the 2000 customers, there was a marginal bias towards female customers in the dataset accounting for 56%, while males accounted for 44%[1]. In terms of education level, 88% of the customer base had a university degree or higher with the largest category of customers coming under 'graduate' at 45% and the smallest was 'basic' at 2.5%[2] .

The numerical variables included age, income, spending score and loyalty points. The average age of customers was 39 years old with youngest recorded as 17 years old, while the oldest was 72[3]. Income distribution was slightly skewed to the right, the average customer earning £48,000[4].

Using a correlation matrix, we were able to highlight a relatively strong correlation between income (0.62) and spending score (0.67) with loyalty points, while there was a weak negative correlation between age and spending score (-0.22)[5].

[1] Distribution of gender
[2] Distribution of education
[3] Distribution of age
[4] Distribution of income
[5] Correlation Matrix

Loyalty points predictor

In order to assess the impact of spending score, income and age on loyalty points, three different types of regression analysis were developed and tested in order to find a model that accurately estimates loyalty points.

Linear regression of the independent variables (income, spending score and age) showed that income and spending score accounted for 32%[6] and 46%[7] respectively of the variance in loyalty points. Essentially this means that as customer incomes and spending score increases, so do their loyalty points. Scatterplots visualised the strength and direction of the relationships with loyalty points to assess the accuracy and it was evident that conditional heteroscedasticity was present in the models. The means that as the independent variables increased, the accuracy of the model fell and is consequently unreliable. The analysis showed that age[8] was not a good estimator for loyalty points and was accordingly disregarded in subsequent analysis.

Using income and spending score as independent variables in multiple linear regression resulted in 83%[9] accuracy in predicting loyalty points, however, the problem of heteroskedascity remained. We transformed the variables but It was concluded that both, linear and multiple linear regression were unreliable models.

Decision tree regression provided the most accurate predictor of loyalty points. The decision tree predicted customer lotalty points based on their income and spending score. Starting from the top, it splits customers into groups by asking yes/no questions such as whether their spend score is above or below a certain threshold. Each path leads to a predicted loyalty point value. With a very low error rate and 96% accuracy in predicting loyalty points[10] it was the most reliable model observed.

Customer Segments

K-means clustering was used to segment the customer base into distinct groups. It was evident once visualised using a scatter plot there were 5 distinct groups based on spending score and income[11]. This was also confirmed using the Elbow[12] and Silhouette[13] method to determine the optimal number of clusters. Each cluster was labelled accordingly to help inform targeted strategies for specific market segments, for example implementing loyalty programs for high income but low spend customers[14].

---

[6] Income – Linear Regression
[7] Spending Score – Linear Regression
[8] Age – Linear Regression
[9] Multiple Regression – Income & Spending Score
[10] Decision Tree Regression
[11] Scatterplot – income and spending score
[12] Elbow Method
[13] Silhouette
[14] Customer Clusters

Product Reviews

Using two Natural Language Processing (NLP) techniques, we were able to analyse the customer reviews to identify the most frequently used words and calculate sentiment and subjectivity scores. Word clouds[15] and histograms[16] visualise the results of this analysis. Overall, the analysis showed that customer reviews were overwhelmingly positive[17]. The analysis found only one review with a -1 polarity score, while there were 22 reviews with the maximum score of +1. Furthermore, 1614 reviews scored between 0 and 1, while only 316 scored between -1 and 0.

Conclusion

The analysis highlighted several key takeaways.

Firstly, spending score and income are the most significant drivers of loyalty points, with the decision tree regression emerging as the most accurate predictive model, achieving 96% accuracy. It should be noted, decision trees are prone to overfitting and although accuracy performance is high this may not hold up on unseen data.

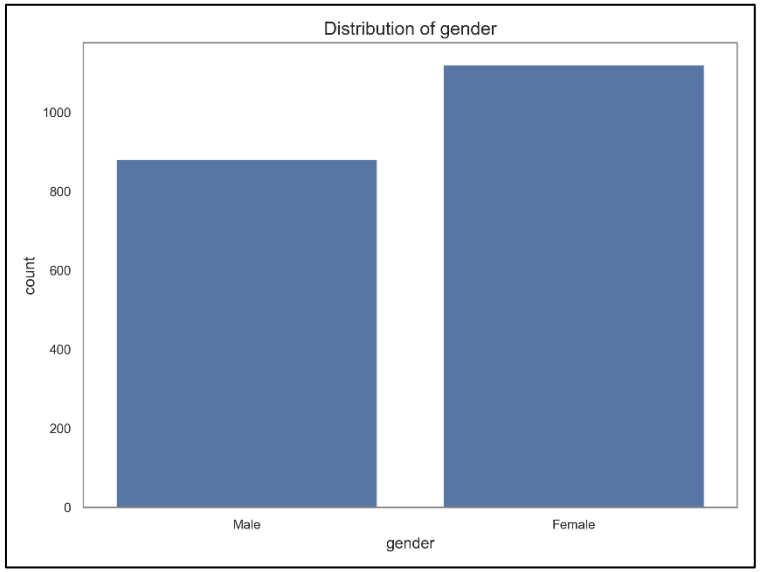
Secondly, customer segmentation revealed five distinct customer groups, enabling more targeted and effective marketing strategies. Further investigation should be prioritised on the 'low income/high spend' segment to understand the drivers of these customers. To improve overall sales for the business, Turtle Games should target the 'high income/low spend' segment with incentives to boost spending. Analysing the impact of previous marketing campaigns on loyalty points may provide insights into how future campaigns could be more focused.

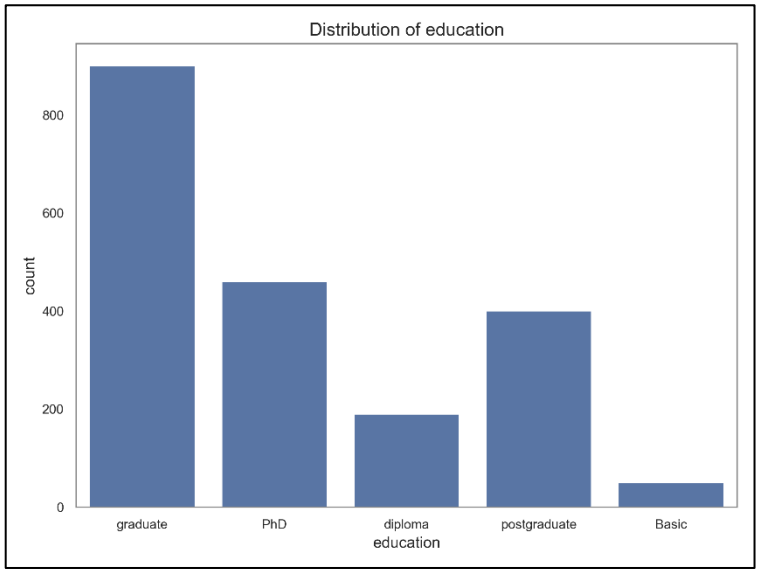
Sentiment analysis of customer reviews indicated a predominance of positive feedback. While this is often observed in similar datasets, it may affect the representativeness of the data. The information collected for this analysis might not reflect all customer experiences, as those with negative feedback may be less likely to submit reviews, resulting in potential data imbalance. Further investigation into the reasons for this bias is recommended, and the use of open-ended questions such as "How can we improve?" may help identify areas not currently addressed.

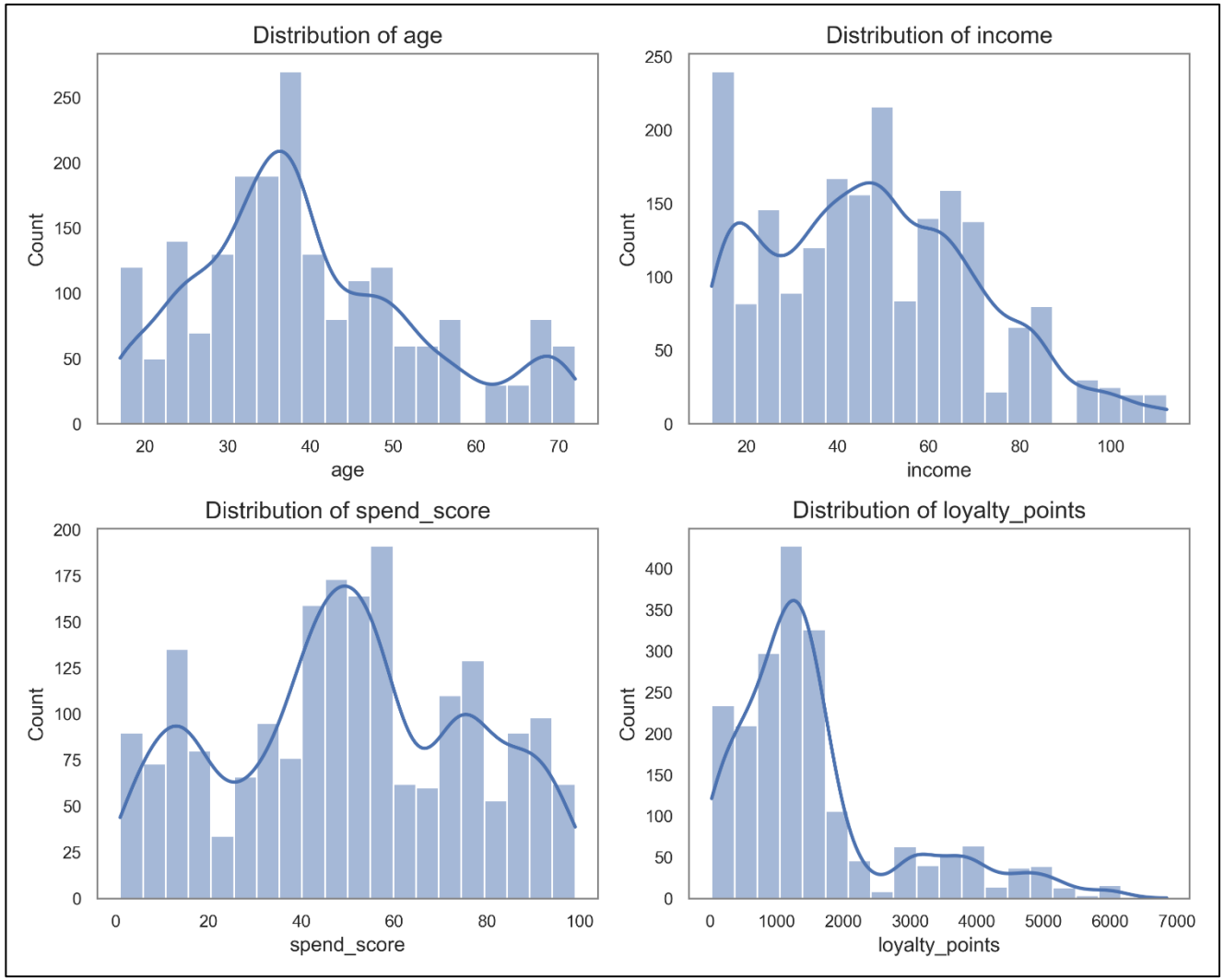
[15] Word Clouds
[16] Histrogram of Sentiment Score Polarity
[17] VADER Sentiment Analysis

Appendix

Exploratory Data Analysis

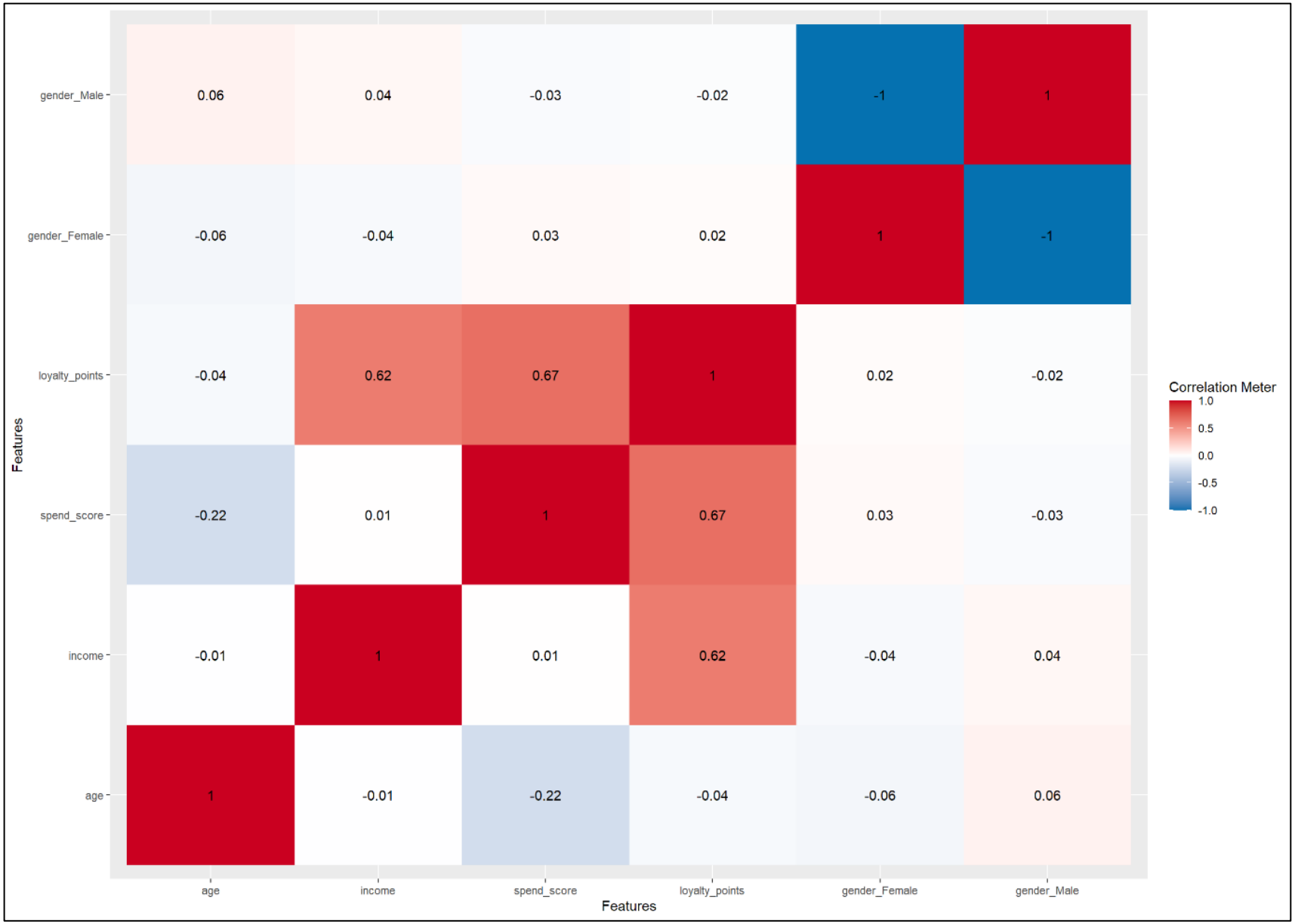






Peter Mackin

Assignment 3: Predicting Future Outcomes
Correlation Matrix

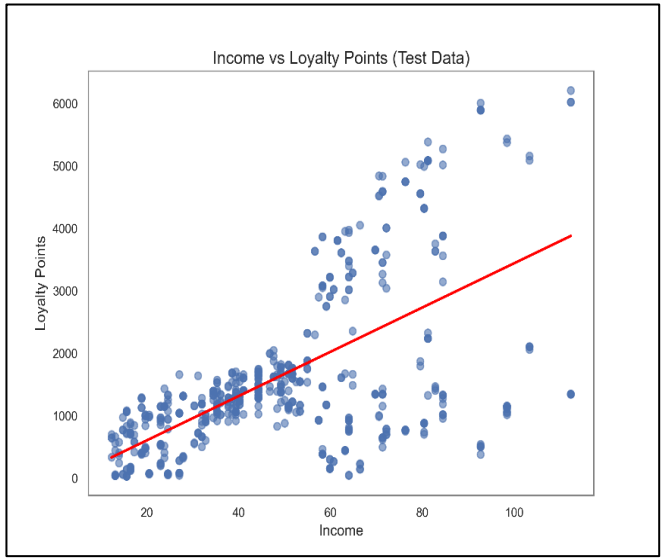


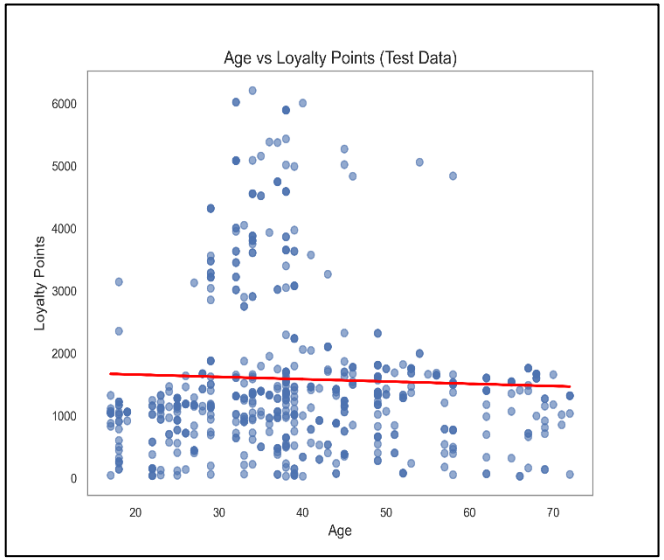
Peter Mackin

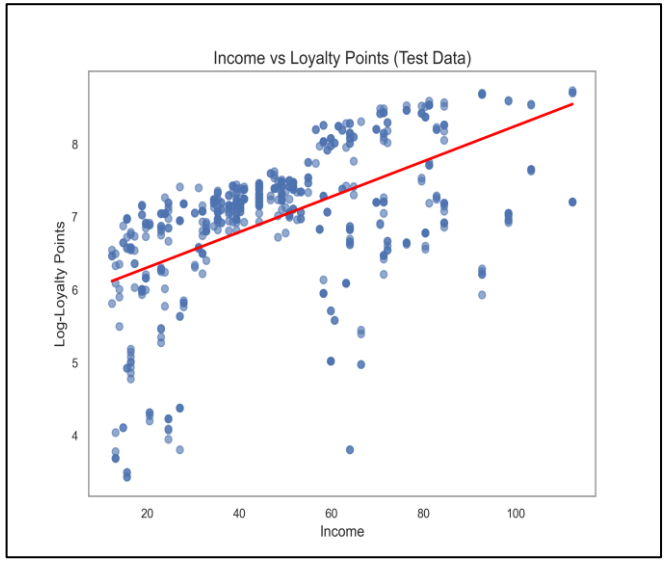
Assignment 3: Predicting Future Outcomes

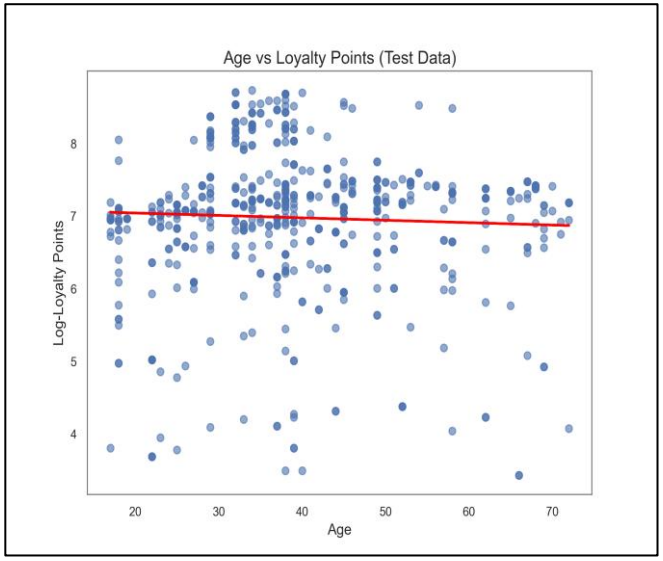
Linear Regression

Top row – income, spending score and age vs loyalty points; Bottom row – income, spending score and age vs Log Loyalty points







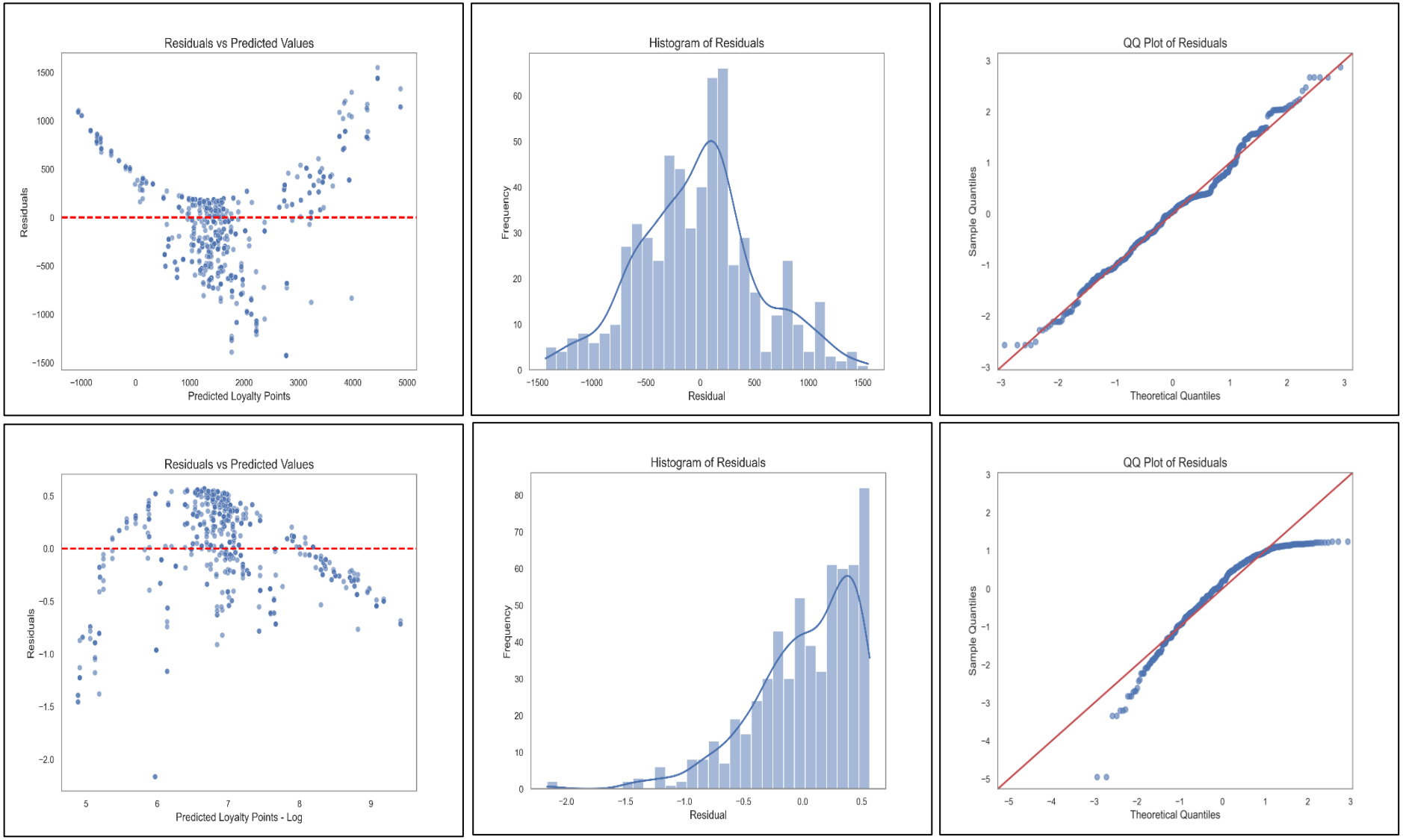


Peter Mackin

Assignment 3: Predicting Future Outcomes

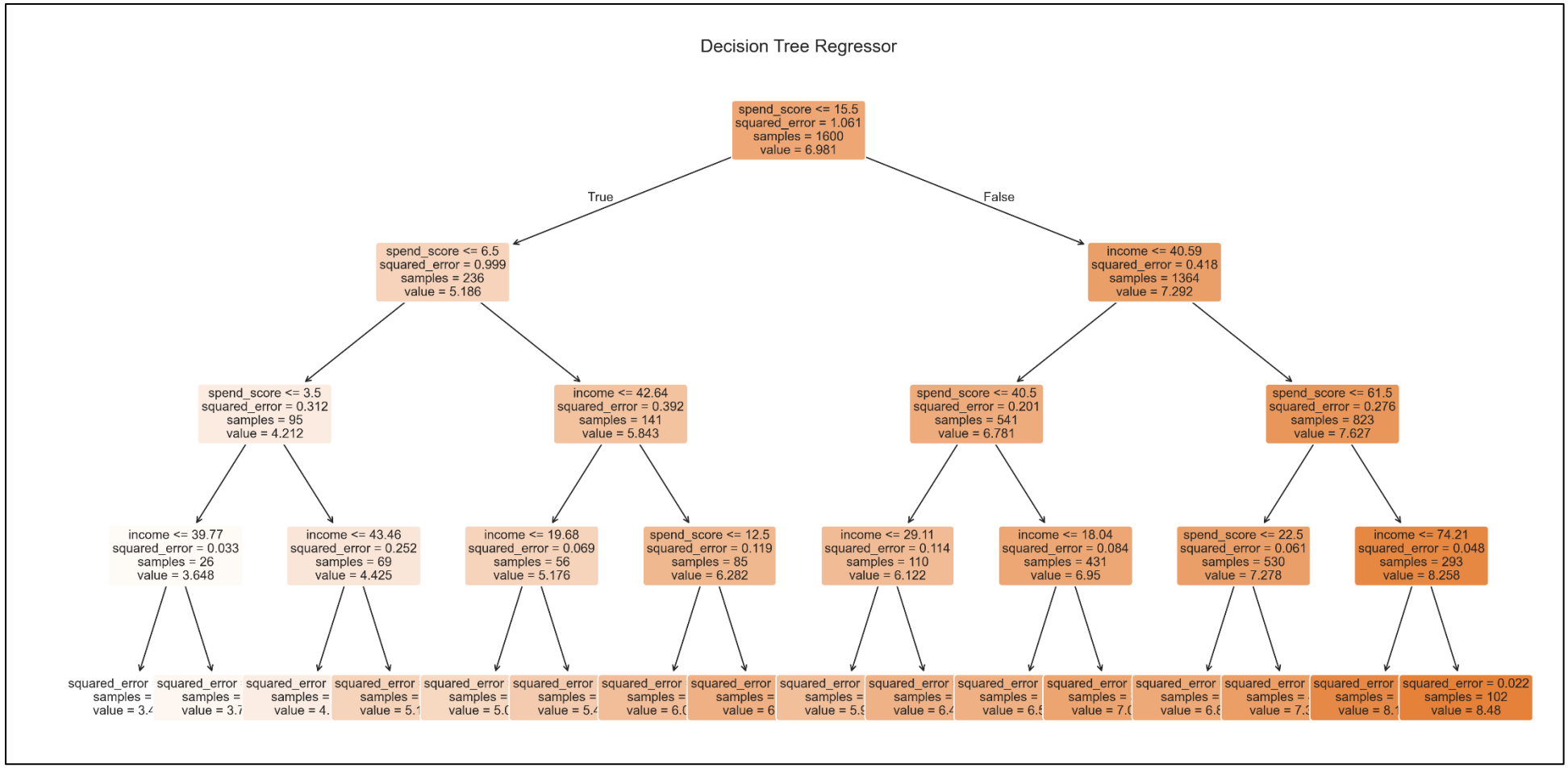
Multiple Linear Regression

Top row – income and spending score vs loyalty points; Bottom row – income and spending score vs log loyalty points



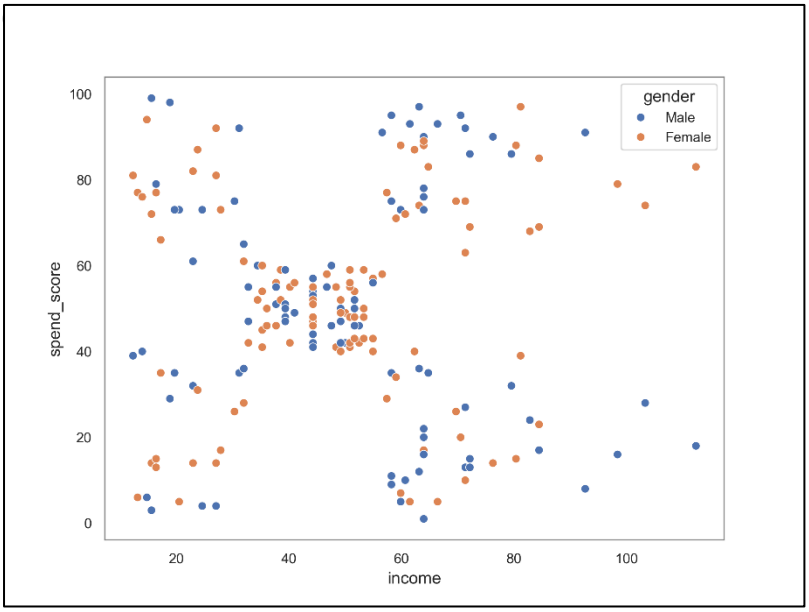
Peter Mackin

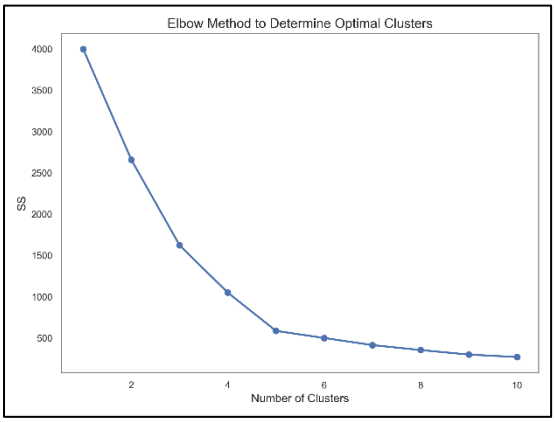
Decision Tree Regression

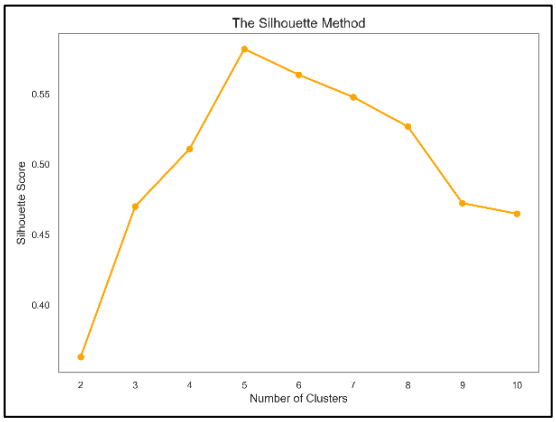


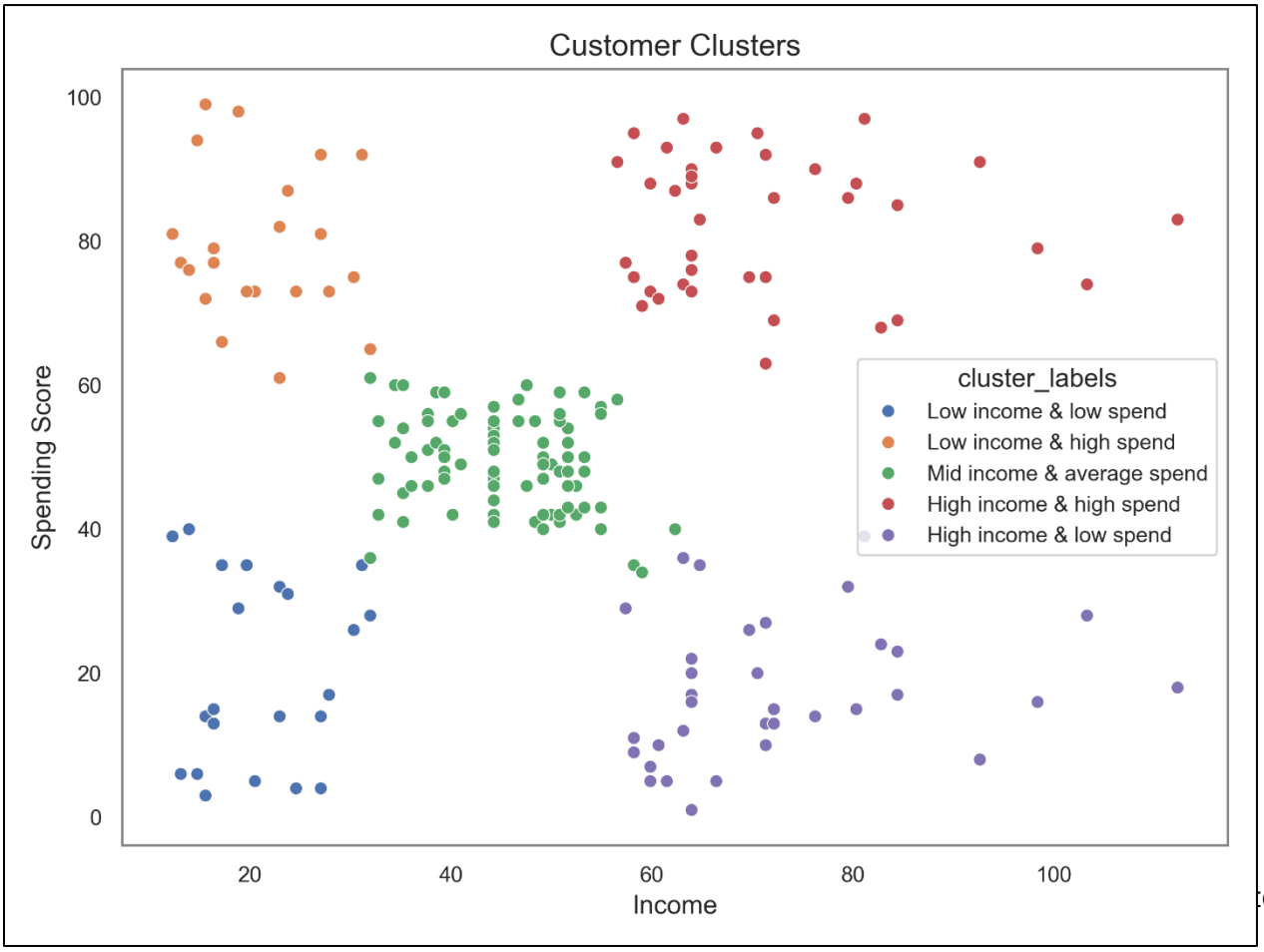
Peter Mackin

Assignment 3: Predicting Future Outcomes









er Mackin

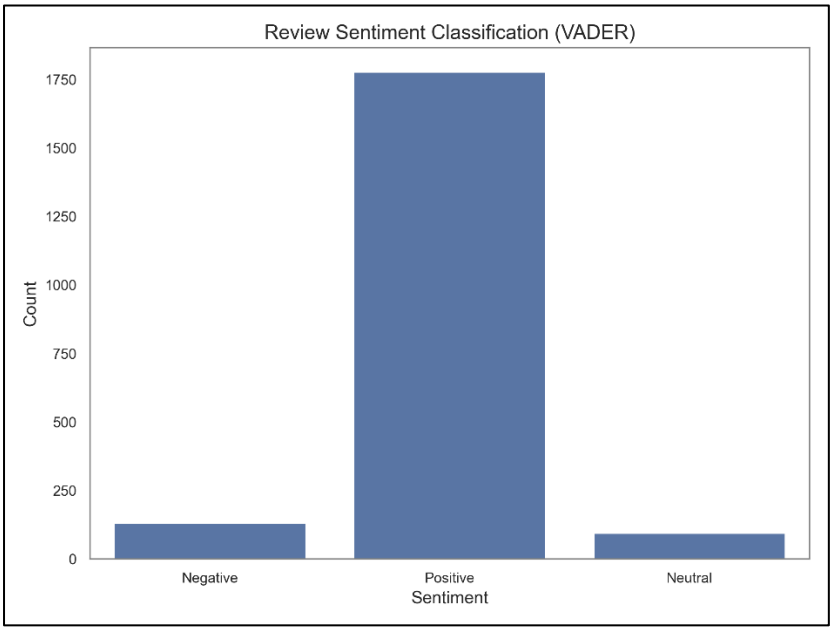
Assignment 3: Predicting Future Outcomes

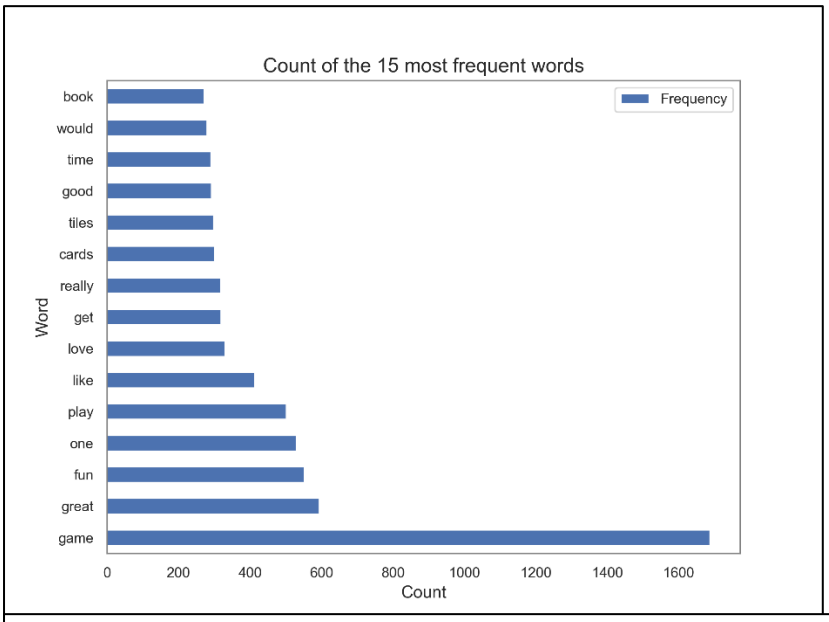
Word Cloud

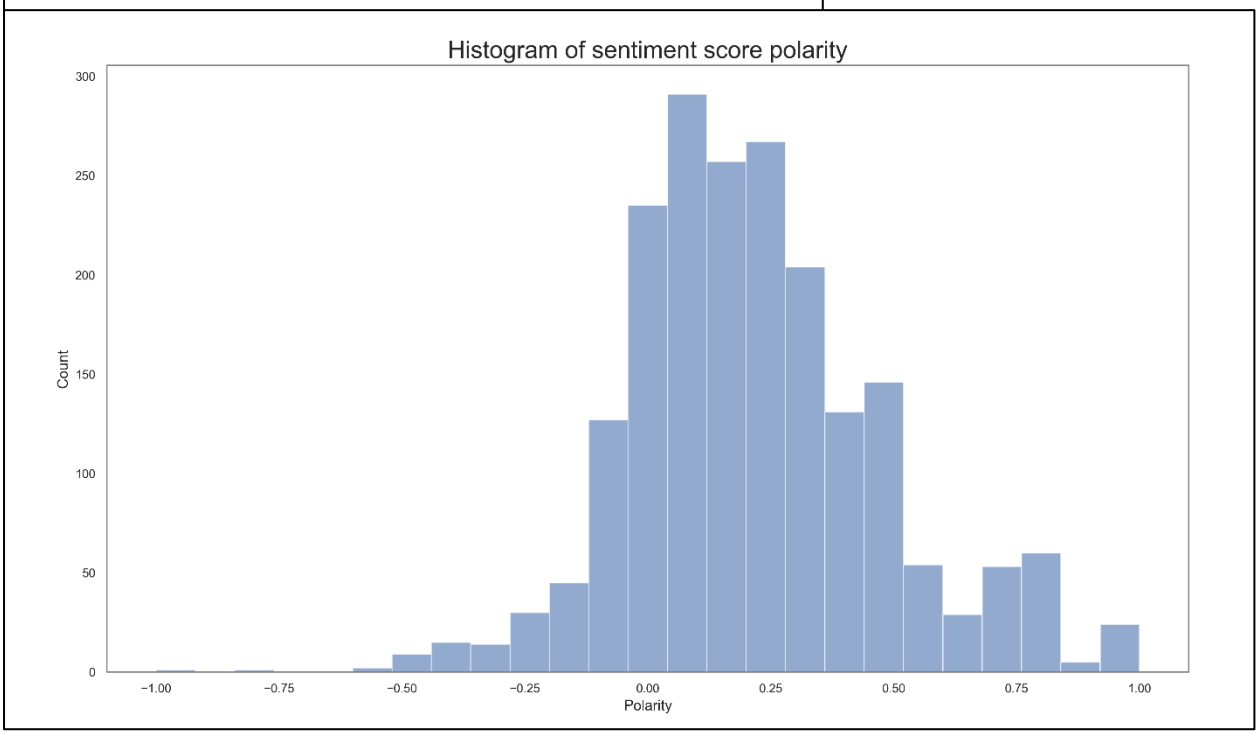
Peter Mackin

Assignment 3: Predicting Future Outcomes

Sentiment Analysis







Peter Mackin